JEE–MAIN EXAMINATION – APRIL 2026

(HELD ON MONDAY 06th APRIL 2026) **TIME : 3:00 PM TO 6:00 PM**

MATHEMATICS — TEST PAPER WITH SOLUTION

SECTION-A

1. Let $f : R \to R$ be defined as $f(x) = \frac{2x^2 - 3x + 2}{3x^2 + x + 3}$.

Then f is :

(1) both one-one and onto

(2) one-one but not onto

(3) onto but not one-one

(4) neither one-one nor onto

Sol. $\frac{2x^2 - 3x + 2}{3x^2 + x + 3} = y$

$(3y - 2)x^2 + (y + 3)x + (3y - 2) = 0$

$D \geq 0$

$(7y - 1)(5y - 7) \leq 0$

$y \in \left[\frac{1}{7}, \frac{7}{5}\right]$

Hence f(x) is into and f(x) is many-one because f(x) is non-monotonic

2. Consider the quadratic equation $(n^2 - 2n + 2)x^2 - 3x + (n^2 - 2n + 2)^2 = 0$, $n \in R$. Let α be the minimum value of the product of its roots and β be the maximum value of the sum of its roots. Then the sum of the first six terms of the G.P., whose first term is α and the common ratio is $\frac{\alpha}{\beta}$, is :

(1) $\frac{61}{37}$ (2) $\frac{121}{81}$

(3) $\frac{364}{243}$ (4) $\frac{1093}{729}$

Sol. $\alpha = n^2 - 2n + 2$

$\alpha = (n - 1)^2 + 1$

$\therefore$ minimum value of α is 1

Similarly $\beta = \frac{3}{(n-1)^2 + 1}$

$\therefore$ maximum value of β is 3

required G.P. is $1, \frac{1}{3}, \frac{1}{3^2}, \ldots$

$$S_6 = \frac{1\left(1 - \left(\frac{1}{3}\right)^6\right)}{1 - \frac{1}{3}} = \frac{3}{2}\left[1 - \left(\frac{1}{3}\right)^6\right] = \frac{364}{243}$$

3. Let $S = \left\{z \in \mathbb{C} : z^2 + \sqrt{6}iz - 3 = 0\right\}$. Then $\sum_{z \in S} z^8$ is equal to :

(1) 162 (2) 184

(3) 262 (4) 324

Sol. $z = \frac{-\sqrt{6}i \pm \sqrt{-6 + 12}}{2}$

$= \frac{-\sqrt{6}i \pm \sqrt{6}}{2}$

$= \frac{\sqrt{6}}{2}(\pm 1 - i)$

$\alpha = \frac{-\sqrt{6}}{2}(1 + i), \beta = \frac{\sqrt{6}}{2}(1 - i)$

$\alpha = \frac{-\sqrt{6}}{2} \times \sqrt{2}e^{i\pi/4}, \beta = +\frac{\sqrt{6}}{2} \times \sqrt{2}^{-i\pi/4}$

$\alpha = -\sqrt{3}e^{i\pi/4}, \beta = +\sqrt{3}e^{-i\pi/4}$

$\alpha^8 + \beta^8 = 81.e^{i2\pi} + 81.e^{i(-2\pi)}$

$= 81\ (1 + 1) = 162$

4. The sum of all possible values of $\theta \in [0, 2\pi]$, for which the system of equations :

$x\cos 3\theta - 8y - 12z = 0$

$x\cos 2\theta + 3y + 3z = 0$

$x + y + 3z = 0$

has a non-trivial solution, is equal to :

(1) π (2) 2π

(3) 3π (4) 4π

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Sol. $\begin{vmatrix} \cos 3\theta & -8 & -12 \\ \cos 2\theta & 3 & 3 \\ 1 & 1 & 3 \end{vmatrix} = 0$

$\begin{vmatrix} \cos 3\theta & -8 & -4 \\ \cos 2\theta & 3 & 1 \\ 1 & 1 & 1 \end{vmatrix} = 0$

$(C_1 \to C_1 - C_2)$ & $C_2 \to C_2 - C_3$

$\begin{vmatrix} \cos 3\theta + 8 & -4 & -3 \\ \cos 2\theta - 3 & 2 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 0$

$2\cos 3\theta + 16 + 4\cos 2\theta - 12 = 0$

$(4\cos^3\theta - 3\cos\theta) + 2(2\cos^2\theta - 1) + 2 = 0$

$4\cos^3\theta + 4\cos^2\theta - 3\cos\theta = 0$

$\cos\theta\,(4\cos^2\theta + 4\cos\theta - 3) = 0$

$\cos\theta\,(2\cos\theta + 3)\,(2\cos\theta - 1) = 0$

$\cos\theta = 0,\ \frac{1}{2},\ -\frac{3}{2}$ (rejected)

$\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{3}, \frac{5\pi}{3}$

Sum $= 4\pi$

5. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 9 & 3 & 1 \end{bmatrix}$ and $B = [b_{ij}]$, $1 \le i, j \le 3$.

If $B = A^{99} - I$, then the value of $\frac{b_{31} - b_{21}}{b_{32}}$ is :

(1) 99 (2) 199

(3) 149 (4) 159

Sol. $A = P + I$

$A = \begin{bmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 9 & 3 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$A = P + I$ where $P^3 = 0$

$A^{99} = I + 99P + {}^{99}C_2P^2$

$A^{99} - I = 99\begin{bmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 9 & 3 & 0 \end{bmatrix} + {}^{99}C_2\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 9 & 0 & 0 \end{bmatrix}$

$= \begin{bmatrix} 0 & 0 & 0 \\ 99\times 3 & 0 & 0 \\ 44550 & 99\times 3 & 0 \end{bmatrix}$

$b_{31} = 44550,\ b_{21} = 99 \times 3,\ b_{32} = 99 \times 3$

$\frac{44550 - 297}{297} = 149$

6. The sum $1 + \frac{1}{2}(1^2 + 2^2) + \frac{1}{3}(1^2 + 2^2 + 3^2) + \ldots\ldots$ upto 10 terms is equal to :

(1) 130 (2) 155

(3) $\frac{315}{2}$ (4) $\frac{325}{2}$

Sol. $T_n = \frac{1}{n}(1^2 + 2^2 + \ldots + n^2)$

$T_n = \frac{n(n+1)(2n+1)}{6n}$

$T_n = \frac{1}{6}[2n^2 + 3n + 1]$

$S_n = \Sigma T_n = \frac{1}{6}[2\Sigma n^2 + 3\Sigma n + \Sigma 1]$

$\Sigma T_n = \frac{1}{6}\left[2.\frac{n(n+1)(2n+1)}{6} + \frac{3n(n+1)}{2} + n\right]$

$S_{10} = \frac{1}{6}\left[\frac{2\times 10\times 11\times 21}{6} + \frac{3}{2}\times 10\times 11 + 10\right]$

$= \frac{10}{6}\left[77 + \frac{33}{2} + 1\right] = \frac{315}{2}$

7. A building has ground floor and 10 more floors. Nine persons enter in a lift at the ground floor. The lift goes up to the 10th floor. The number of ways, in which any 4 persons exit at a floor and the remaining 5 persons exit at a different floor, if the lift does not stop at the first and the second floors, is equal to :

(1) 2184 (2) 3064

(3) 7056 (4) 11340

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Sol. Form group of 4 and 5 person $\frac{9!}{4!5!}=126$

choose two floor out of 8 floor ${}^{8}C_{2}=28$

Total ways in which 4 & 5 persons can leave at different floor $126\times28\times2=7056$

8. Let the mean and the variance of seven observations 2, 4, α, 8, β, 12, 14, $\alpha<\beta$, be 8 and 16 respectively. Then the quadratic equation whose roots are $3\alpha+2$ and $2\beta+1$ is :

(1) $x^2-35x+306=0$ (2) $x^2-41x+420=0$

(3) $x^2-45x+506=0$ (4) $x^2-37x+342=0$

Sol. $\frac{2+4+\alpha+\beta+8+12+14}{7}=8$

$\alpha+\beta=56-40=16\Rightarrow\alpha+\beta=16$ .....(i)

$\text{Variance}=\frac{\sum x_i^2}{n}-\left(\frac{\sum x_i}{n}\right)^2$

$16=\frac{2^2+4^2+\alpha^2+\beta^2+8^2+12^2+14^2}{7}-64$

$560=\alpha^2+\beta^2+424$

$\Rightarrow\alpha^2+\beta^2=136$ ....(ii)

From (i) & (ii) $\alpha=6$ & $\beta=10$

$x_1=3\alpha+2=20$; $x_2=2\beta+1=21$

Q.E. : $x^2-41x+420=0$

9. A bag contains 6 blue and 6 green balls. Pairs of balls are drawn without replacement until the bag is empty. The probability that each drawn pair consists of one blue and one green ball is :

(1) $\frac{63}{925}$ (2) $\frac{17}{231}$

(3) $\frac{16}{231}$ (4) $\frac{64}{925}$

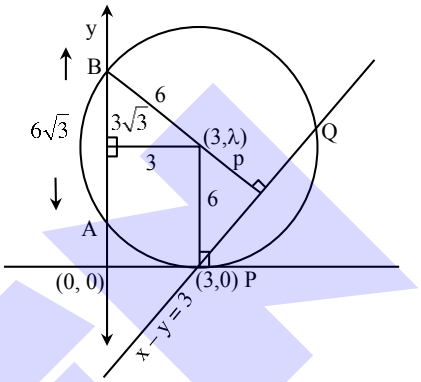
Sol. $\text{Probability}=\frac{\lfloor6\times\lfloor6}{\left(\frac{\lfloor12}{(\lfloor2)^6\lfloor6}\right)\times\lfloor6}=\frac{16}{231}$

10. Let C be a circle having centre in the first quadrant and touching the x-axis at a distance 3 units from the origin. If the circle C has an intercept of length $6\sqrt{3}$ on y-axis, then the length of the chord of the circle C on the line $x-y=3$ is :

(1) 8 (2) 6

(3) $6\sqrt{2}$ (4) $8\sqrt{2}$

Sol. Centre of circle is (3, 6) and radius = 6 units.



$PQ=2\sqrt{6^2-p^2}$ $\therefore p=\frac{6}{\sqrt{2}}$

$p=\frac{|3-6-3|}{\sqrt{2}}$

$PQ=2\sqrt{36-\frac{36}{2}}$

$=2\times\frac{6}{\sqrt{2}}=6\sqrt{2}$

11. The eccentricity of an ellipse E with centre at the origin O is $\frac{\sqrt{3}}{2}$ and its directrices are $x=\pm\frac{4\sqrt{6}}{3}$.

Let H : $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ be a hyperbola whose eccentricity is equal to the length of semi-major axis of E, and whose length of latus rectum is equal to the length of minor axis of E. Then the distance between the foci of H is :

(1) $\frac{4\sqrt{2}}{\sqrt{7}}$ (2) $\frac{4\sqrt{2}}{7}$

(3) $\frac{4}{\sqrt{7}}$ (4) $\frac{8}{7}$

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Sol. $e = \frac{\sqrt{3}}{2}$

$\frac{a}{e} = \frac{4\sqrt{6}}{3}$

$a = 2\sqrt{2}$

$(ae)^2 = a^2 - b^2$

$b^2 = 2$

so E : $\frac{x^2}{8} + \frac{y^2}{2} = 1$

Now $e_H = 2\sqrt{2} \Rightarrow b^2 = 7a^2$

$\frac{2b^2}{a} = 2\sqrt{2} \Rightarrow a = \frac{\sqrt{2}}{7}$

$(ae_H)^2 = a^2 + b^2 = 8a^2 = \frac{16}{49}$

$ae_H = \frac{4}{7}$

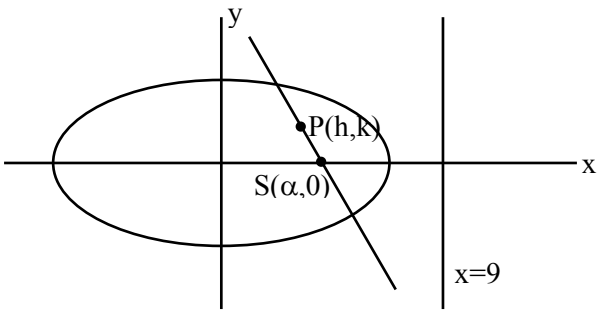
Distance between focii $= 2ae_H = \frac{8}{7}$

12. Let x = 9 be a directrix of an ellipse E, whose centre is at the origin and eccentricity is $\frac{1}{3}$. Let P (α, 0), $\alpha > 0$, be a focus of E and AB be a chord passing through P. Then the locus of the mid point of AB is :

(1) $9y^2 = 8x(1 - x)$ (2) $3y^2 = 4x(1 - x)$

(3) $9y^2 = 8x(x - 1)$ (4) $3y^2 = 4x(x - 1)$

Sol.



$\frac{a}{e} = 9 \Rightarrow a = 3$

$b^2 = a^2(1 - e^2) = 9\left(1 - \frac{1}{9}\right) = 8$

$\frac{x^2}{9} + \frac{y^2}{8} = 1$

$S(\alpha, 0) \equiv S(1, 0)$

$T = S_1 \Rightarrow \frac{hx}{9} + \frac{ky}{8} = \frac{h^2}{9} + \frac{k^2}{8}$

$(1, 0) \Rightarrow \frac{h}{9} + 0 = \frac{h^2}{9} + \frac{k^2}{8}$

$h = h^2 + \frac{9}{8}k^2$

$9y^2 = 8x(1 - x)$

13. If $\sin\left(\tan^{-1}(x\sqrt{2})\right) = \cot\left(\sin^{-1}\sqrt{1-x^2}\right)$, $x \in (0, 1)$, then the value of x is :

(1) $\frac{1}{2}$ (2) $\frac{1}{3}$

(3) $\frac{2}{3}$ (4) $\frac{5}{8}$

Sol. $\frac{x\sqrt{2}}{\sqrt{2x^2+1}} = \frac{1}{\sqrt{1-x^2}}$

$2(1 - x^2) = (2x^2 + 1)$

$2 - 2x^2 = 2x^2 + 1$

$x^2 = \frac{1}{4}$

$x = \frac{1}{2}$

14. The shortest distance between the lines $\frac{x-4}{1} = \frac{y-3}{2} = \frac{z-2}{-3}$ and $\frac{x+2}{2} = \frac{y-6}{4} = \frac{z-5}{-5}$ is:

(1) $\frac{5\sqrt{6}}{6}$ (2) $2\sqrt{5}$

(3) $3\sqrt{5}$ (4) $4\sqrt{5}$

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Sol. $SD = \dfrac{\begin{vmatrix} 4-(-2) & 3-6 & 2-5 \\ 1 & 2 & -3 \\ 2 & 4 & -5 \end{vmatrix}}{\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -3 \\ 2 & 4 & -5 \end{vmatrix}}$

$$SD = \frac{|6(-10+12)-(-3)(-5+6)-3(4-4)|}{|2\hat{i}-\hat{j}|}$$

$$SD = \left|\frac{12+3}{\sqrt{5}}\right|$$

$$= \frac{15}{\sqrt{5}}$$

$$SD = 3\sqrt{5}$$

15. Let $\vec{a}=2\hat{i}+3\hat{j}+3\hat{k}$ and $\vec{b}=6\hat{i}+3\hat{j}+3\hat{k}$. Then the square of the area of the triangle with adjacent sides determined by the vectors $(2\vec{a}+3\vec{b})$ and $(\vec{a}-\vec{b})$ is :

(1) 450

(2) 900

(3) 1800

(4) 2400

Sol. $2\vec{a}+3\vec{b}=2(2\hat{i}+3\hat{j}+3\hat{k})+3(6\hat{i}+3\hat{j}+3\hat{k})$

$\Rightarrow 22\hat{i}+15\hat{j}+15\hat{k}$

$\vec{a}-\vec{b}=-4\hat{i}$

$$\text{Area} = \frac{1}{2}\left|(2\vec{a}+3\vec{b})\times(\vec{a}-\vec{b})\right|$$

$$= \frac{1}{2}\left|-60\hat{j}+60\hat{k}\right| = \frac{1}{2}\sqrt{(60)^2\times 2}$$

$$A = \frac{60}{\sqrt{2}}$$

Square of area is $A^2 = 1800$

16. Let $\lim\limits_{x\to 2}\dfrac{(\tan(x-2))\left(rx^2+(p-2)x-2p\right)}{(x-2)^2} = 5$ for some r, p $\in$ R. If the set of all possible values of q, such that the roots of the equation $rx^2 - px + q = 0$ lie in (0, 2), be the interval $(\alpha, \beta]$, then $4(\alpha + \beta)$ equals.

(1) 11 (2) 13

(3) 17 (4) 21

Sol. $\lim\limits_{x\to 2}\dfrac{\tan(x-2)}{x-2}\cdot\dfrac{[rx^2+(p-2)x-2p]}{x-2}=5$

$\Rightarrow \lim\limits_{x\to 2}\dfrac{rx^2-2x+p(x-2)}{x-2}=5$

$\because D^r \to 0 \therefore N^r \to 0 \Rightarrow r = 1$

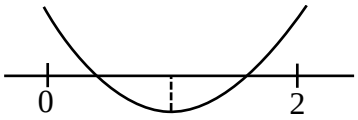
$\Rightarrow \lim\limits_{x\to 2}\dfrac{x(x-2)+p(x-2)}{x-2}=5$

$\Rightarrow 2 + p = 5$

$\Rightarrow p = 3$

Now, quadratic equation is $x^2 - 3x + q = 0$

its both roots lie in (0, 2)



$$D \geq 0, \quad 0 < \frac{-b}{2a} < 2, \quad f(0) > 0, \quad f(2) > 0$$

$\Rightarrow 9 - 4q \geq 0$ and $q > 2$

$\Rightarrow q \in \left(2, \dfrac{9}{4}\right]$

$\alpha = 2, \beta = \dfrac{9}{4} \Rightarrow \left(2+\dfrac{9}{4}\right) = 17$

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17. Let $A=\begin{vmatrix}1 & 3 & -1\\ 2 & 1 & \alpha\\ 0 & 1 & -1\end{vmatrix}$ be a singular matrix. Let $f(x)=\int_0^x (t^2+2t+3)dt$, $x \in [1, \alpha]$. If M and m are respectively the maximum and the minimum values of f in $[1, \alpha]$ then $3(M-m)$ is equal to :

(1) 64 (2) 68

(3) 72 (4) 76

Sol. $|A| = 0$

$\Rightarrow 1.(-1-\alpha)-3(-2-0)-1(2-0)=0$

$\Rightarrow \alpha = 3$

$f(x)=\int_0^x (t^2+2t+3)dt \; ; x \in [1, 3]$

$f(x)=\frac{x^3}{3}+x^2+3x$

$f'(x) = x^2 + 2x + 3$ where $D < 0$

$\Rightarrow$ f(x) is strictly increasing

$M = f(3) = 9 + 9 + 9 = 27$

$m = f(1)=\frac{1}{3}+1+3=\frac{13}{3}$

$3(M-m)=3\left(27-\frac{13}{3}\right)=68$

18. Let $f : R \to R$ be such that $f(xy) = f(x)f(y)$, for all $x, y \in R$ and $f(0) \neq 0$. Let $g: [1, \infty) \to R$ be a differentiable function such that

$x^2g(x)=\int_1^x (t^2f(t)-tg(t))dt$

Then g(2) is equal to :

(1) $\frac{13}{8}$ (2) $\frac{11}{16}$

(3) $\frac{15}{32}$ (4) $\frac{17}{64}$

Sol. $\because f(xy) = f(x)f(y)$

$\therefore f(x) = x^n$ or $f(x) = k$

But $f(0) \neq 0 \Rightarrow f(x) \neq x^n$

For $f(x) = k \Rightarrow k = k.k \Rightarrow k = 1 (\because f(0) \neq 0)$

$\Rightarrow f(x) = 1$

$x^2g(x)=\int_1^x (t^2(1)-tg(t)).dt$

diff w/r to x

$\Rightarrow x^2g'(x) + g(x) . 2x = x^2 - xg(x)$

$\Rightarrow xg'(x) + 3g(x) = x$

$\Rightarrow g'(x)+\frac{3}{x}g(x)=1$

$IF = e^{\int \frac{3}{x}dx} = e^{3\ln(x)} = x^3$

Solution of DE :

$\Rightarrow g(x) . x^3 = \int x^3dx$

$\Rightarrow x^3.g(x)=\frac{x^4}{4}+C$

$\Rightarrow C=-\frac{1}{4}$ $\{\because g(1) = 0\}$

$g(x) . x^3=\frac{x^4}{4}-\frac{1}{4}$

Put x = 2

(8) $g(2)=4-\frac{1}{4}$

$g(2)=\frac{15}{32}$.

19. The area of the region $\{(x, y) : x^2 - 8x \le y \le -x\}$ is:

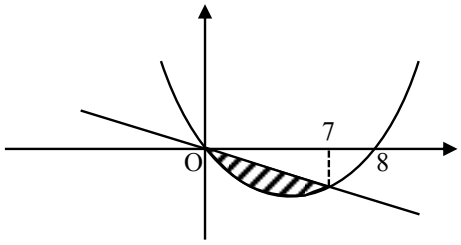
(1) $\frac{343}{6}$ (2) $\frac{637}{6}$

(3) $\frac{437}{6}$ (4) $\frac{523}{6}$

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Sol. Required area $= \int_0^7 (7x - x^2)\,dx$

$$= \left[\frac{7x^2}{2} - \frac{x^3}{3}\right]_0^7 = \frac{343}{6}$$



20. The value of the integral $\int_{-1}^{1}\left(\frac{x^3+|x|+1}{x^2+2|x|+1}\right)dx$ is equal to

(1) $3\log_e 2$ (2) $2\log_e 2$

(3) $5\log_e 3$ (4) $3\log_e 3$

Sol. $I = \int_{-1}^{1} \frac{x^3+|x|+1}{x^2+2|x|+1}.dx$

$I = \int_{-1}^{1} \frac{x^3}{x^2+2|x|+1}.dx + \int_{-1}^{1} \frac{|x|+1}{x^2+2|x|+1}.dx$

$I = 0 + \int_{-1}^{1} \frac{|x|+1}{|x|^2+2|x|+1}.dx$

$2\int_0^1 \frac{1}{x+1}.dx$

$I = 2\left[\ell n|x+1|\right]_0^1$

$I = 2\ell n(2)$

SECTION-B

21. Let $R = \{(x,y) \in N \times N : \log_e (x + y) \le 2\}$. Then the minimum number of elements, required to be added in R to make it a transitive relation, is _____.

Sol. $x + y \le e^2$ ($e^2 \simeq 7.29$)

R = {(1, 1), (1,2), (1, 4), (1,5), (1,6)}
(2, 1), (2,2), (2,3), (2,4), (2,5)
(3,1), (3,2), (3,3), (3,4)
(4,1), (4,2), (4,3)
(5,1), (5,2)
(6,1)

For R to be transitive, we have to add
(6,2), (6,3), (6,4), (6,5), (6,6)
(5,3), (5,4), (5,5), (5,6)
(4,4), (4,5), (4,6)
(3,5), (3,6)
(2,3)

= 15 elements

22. If $(1-x^3)^{10} = \sum_{r=0}^{10} a_r x^r (1-x)^{30-2r}$, then $\frac{9a_9}{a_{10}}$ is equal to _____

Sol. $(1-x)^3 = 1 - x^3 - 3x(1-x)$

$(1-x)^3 = 1 - x^3 - 3x + 3x^2$

$(1-x)^3 + 3x - 3x^2 = 1 - x^3$

$(1-x^3)^{10} = [(1-x)^3 + 3x(1-x)]^{10}$

$\Rightarrow {}^{10}C_r(3x(1-x))^r.((1-x)^3)^{10-r}$

$(1-x^3)^{10} = {}^{10}C_r.3^r.x^r.(1-x)^{30-2r}$

$\therefore a_r = {}^{10}C_r.3^r$

$\frac{a_9}{a_{10}} = \frac{{}^{10}C_9 3^9}{{}^{10}C_{10}.3^{10}}$

$\frac{a_9}{a_{10}} = \frac{10}{3}$

$\therefore \frac{9a_9}{a_{10}} = 30$

23. Let the line $x - y = 4$ intersect the circle $C : (x - 4)^2 + (y + 3)^2 = 9$ at the points Q and R. If $P(\alpha, \beta)$ is a point on C such that PQ = PR, then $(6\alpha + 8\beta)^2$ is equal to _____.

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Sol. $x - y = 4$ … (1)

$(x-4)^2 + (y+3)^2 = 9$ … (2)

(α, β) lies on circle

$\therefore \quad (\alpha-4)^2 + (\beta+3)^2 = 9$ … (3)

eq. (1) & (2) $\Rightarrow y^2 + (y+3)^2 = 9 \Rightarrow 2y^2 + 6y = 0$

$\Rightarrow \quad y = 0, -3$

$\therefore \quad x = 4, 1$

Q(4, 0), R(1, –3)

now

$(PQ)^2 = (PR)^2$

$\Rightarrow \quad (\alpha-4)^2 + \beta^2 = (\alpha-1)^2 + (\beta+3)^2$

$\Rightarrow \alpha + \beta = 1$ … (4)

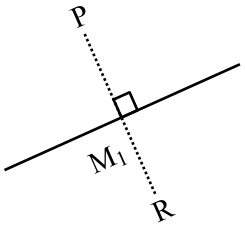
eq. (3) & (4)

$$\Rightarrow \alpha = 4 + \frac{3}{\sqrt{2}}, \ \beta = -3 - \frac{3}{\sqrt{2}}$$

$\therefore (6\alpha + 8\beta)^2 = 18$

24. Let the image of the point P(0, –5, 0) in the line $\frac{x-1}{2} = \frac{y}{1} = \frac{z+1}{-2}$ be the point R and the image of the point $Q\left(0, \frac{-1}{2}, 0\right)$ in the line $\frac{x-1}{-1} = \frac{y+9}{4} = \frac{z+1}{1}$ be the point S. Then the square of the area of the parallelogram PQRS is _____.

Sol. P(0, –5, 0)



$$\frac{x-1}{2} = \frac{y}{1} = \frac{z+1}{-2} = k$$

Let $R(x_1, y_1, z_1)$

$2k + 1, k, -2k - 1$

$(2k+1)(2) + (k+5)(1) + (-2k-1)(-2) = 0$

$\Rightarrow 9k + 9 = 0 \Rightarrow \quad k = -1$

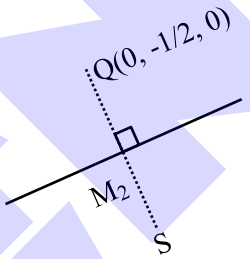
$\therefore \quad M_1 \Rightarrow (-1, -1, 1)$

$R \Rightarrow (-2, 3, 2)$

Q(0, –1/2, 0)

$$\frac{x-1}{-1} = \frac{y+9}{4} = \frac{z+1}{1} = \lambda$$

$(-\lambda + 1, 4\lambda - 9, \lambda - 1)$



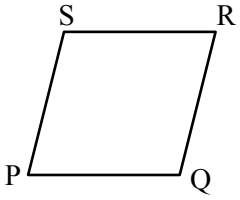
$S(x_2, y_2, z_2)$

$$(-\lambda+1)(-1) + \left(4\lambda - \frac{17}{2}\right)(4) + (\lambda-1)(1) = 0$$

$\Rightarrow 18\lambda - 36 = 0 \Rightarrow \lambda = 2$

$M_2 \Rightarrow (-1, -1, 1)$

$S \Rightarrow (-2, -3/2, 2)$



$PR = \sqrt{4+64+4} = \sqrt{72} = 6\sqrt{2}$

$QS = \sqrt{4+1+4} = \sqrt{9} = 3$

$Ar. = \frac{1}{2} \cdot 6\sqrt{2} \cdot 3 = 9\sqrt{2}$

$(Ar.)^2 = \left(9\sqrt{2}\right)^2 = 162$ Sq.unit

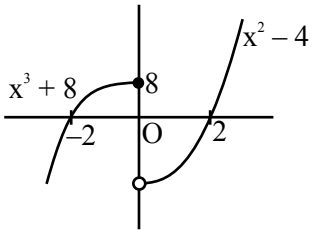
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25. Let $f(x)=\begin{cases}x^3+8 & x<0\\ x^2-4 & x\geq 0\end{cases}$ and

$$g(x)=\begin{cases}(x-8)^{1/3}; & x<0\\ (x+4)^{1/2}; & x\geq 0\end{cases}$$

Then the number of points, where the function gof is discontinuous, is _____.

Sol. $g(f(x))=\begin{cases}(f(x)-8)^{1/3} & f(x)<0\\ (f(x)+4)^{1/2} & f(x)\geq 0\end{cases}$



$$g(f(x))=\begin{cases}(x^3)^{1/3} & , \quad x<-2\\ (x^2-12)^{1/3} & , \quad 0<x<2\\ (x^3+12)^{1/2} & , \quad -2\leq x\leq 0\\ (x^2)^{1/2} & , \quad x\geq 2\end{cases}$$

$$g(f(x))=\begin{cases}x & , \quad x<-2\\ (x^3+12)^{1/2} & , \quad -2\leq x\leq 0\\ (x^2-12)^{1/3} & , \quad 0<x<2\\ x & , \quad x\geq 2\end{cases}$$

Number of points of discontinuity is equal to 3.

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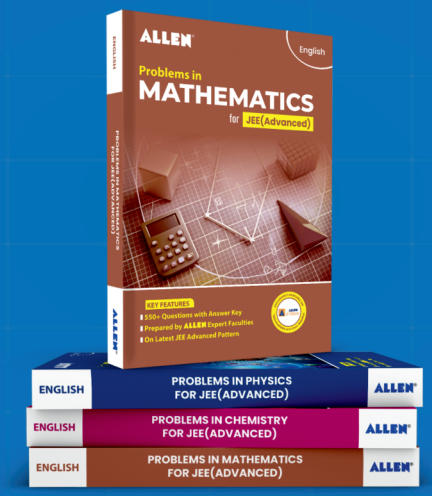


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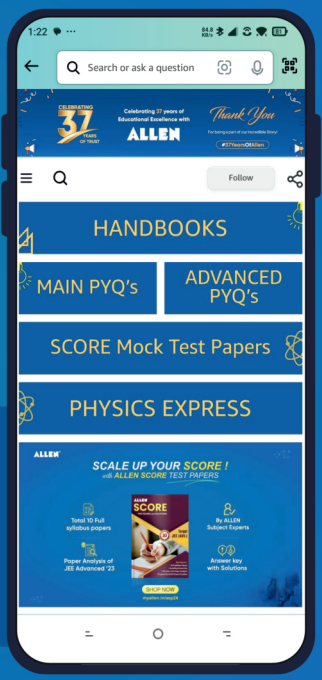
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